

INTRODUCTION TO ROBOTICS & INTELLIGENT CONTROL

End Semester Examination · Full Marks: 100 · Time: 3 Hours

All questions carry marks as indicated. Cheat Sheet provided on last page. Attempt ALL questions.

Instructions: This paper has 4 Sections (A, B, C, D). Attempt all. Marks are **equally weighted** between Electrical (Units 6&7) and Mechanical (Units 1–5). The cheat-sheet on the final page may be detached. Show all working for full credit.

SECTION A — Short Answer & Multiple Choice [20 Marks]

Answer ALL questions. Each carries 2 marks.

Q1. A mobile robot's battery has monitoring points A , B , C with $V_A = 12\text{ V}$, $V_B = 5\text{ V}$, $V_C = -3\text{ V}$. Calculate $V_{AB} + V_{BC} + V_{CA}$. [2 marks]

Solution Q1

$$V_{AB} = V_A - V_B = 12 - 5 = 7\text{ V}$$

$$V_{BC} = V_B - V_C = 5 - (-3) = 8\text{ V}$$

$$V_{CA} = V_C - V_A = -3 - 12 = -15\text{ V}$$

$$\therefore V_{AB} + V_{BC} + V_{CA} = 7 + 8 + (-15) = \boxed{0\text{ V}}$$

This always equals zero by KVL — the sum of potential differences around any closed loop is zero.

Q2. For a revolute joint in a robotic manipulator, which DH parameter is the variable (joint input) while the others remain fixed by design? [2 marks]

Solution Q2

The variable DH parameter for a **revolute joint** is θ_i (the joint angle). The link length a_i , link twist α_i , and joint offset d_i are all fixed geometric constants for revolute joints.

θ_i — the joint angle

Q3. A surgical robot's tool frame is rotated 90° about the z -axis from the base frame. A point at $(1, 0, 0)$ in the tool frame has what coordinates in the base frame? [2 marks]

Solution Q3

Rotation matrix about z by 90° :

$$R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$${}^{\text{base}}\vec{p} = R_z(90^\circ) \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$(0, 1, 0)$ — Answer (a).

Q4. A 7-DOF humanoid arm produces a 6×7 Jacobian matrix. Why can the exact matrix inverse never be computed? [2 marks]

Solution Q4

A 6×7 Jacobian is **non-square** (more columns than rows — the robot is kinematically redundant). Only square matrices have a standard inverse. Instead, the **pseudoinverse** $J^\dagger = J^T(JJ^T)^{-1}$ must be used to find a minimum-norm least-squares solution.

Non-square matrix; exact inverse undefined. Use pseudoinverse.

Q5. State the two key properties that make a 3×3 matrix a valid rotation matrix. [2 marks]

Solution Q5

A matrix R is a valid rotation matrix if and only if:

1. $R^{-1} = R^T$ (orthogonality: columns are mutually orthogonal unit vectors)
2. $\det(R) = +1$ (proper rotation, i.e. no reflection)

These together define the Special Orthogonal Group $SO(3)$.

Q6. A DC motor stalls at $\omega = 0$. Given $V = 110\text{ V}$, $R = 2\ \Omega$, $K = 1.5\text{ Nm/A}$. What is the stall torque? [2 marks]

Solution Q6

At stall, back-EMF = 0, so all voltage drops across armature resistance:

$$I_{\text{stall}} = \frac{V}{R} = \frac{110}{2} = 55\text{ A}$$

$$T_{\text{stall}} = K I_{\text{stall}} = 1.5 \times 55 = \boxed{82.5\text{ Nm}}$$

Q7. Define boundary singularity and interior singularity for a robot manipulator. [2 marks]

Solution Q7

Boundary singularity: occurs at the *edge of the workspace* when the arm is fully extended or fully folded. The end-effector cannot move radially outward/inward.

Interior singularity: occurs *inside the workspace* when two or more joint axes become co-linear or parallel (i.e. the Jacobian loses rank). Certain end-effector velocities require infinite joint velocities — the robot is momentarily unable to move in specific directions.

Q8. Write the 4×4 HTM inverse formula without explicit matrix inversion. [2 marks]

Solution Q8

For $T = \begin{pmatrix} R & \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$, the inverse is:

$$T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

This exploits $R^{-1} = R^T$ (no costly matrix inversion needed) and transforms the translation as $-R^T \vec{p}$.

Q9. What does the manipulator *dexterity* (or manipulability) measure? When is it zero? **[2 marks]**

Solution Q9

Manipulability (dexterity) $w = \sqrt{\det(JJ^T)}$ measures how *freely* the end-effector can move in all directions. A large w means the robot can move equally well in every direction.

$w = 0$ when $\det(J) = 0$, i.e. at a **singular configuration**. For the 2-DOF planar robot this happens when $\sin \theta_2 = 0 \Rightarrow \theta_2 = 0^\circ$ or $\theta_2 = 180^\circ$ (fully extended or folded).

Q10. An n -axis industrial robot requires how many independent DH parameters to describe its full kinematic structure? **[2 marks]**

Solution Q10

Each link-joint pair requires **4 DH parameters** ($\theta_i, d_i, a_i, \alpha_i$). For n joints:

$$\text{Total DH parameters} = \boxed{4n}$$

Answer: (a) $4n$.

SECTION B — Kinematics & Transformations**[30 Marks]**

Answer ALL questions.

B1. Frame $\{B\}$ is obtained from frame $\{A\}$ by rotating 90° about the Z -axis and then translating by $(4, 3, 0)$.

(a) Find the homogeneous transformation matrix ${}^A T_B$. [4]

(b) Find $({}^A T_B)^{-1}$ using the matrix property (no row-reduction). [3]

[7 marks]**Solution B1**

(a) Rotation by 90° about Z :

$$R_z(90^\circ) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \vec{p} = \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix}$$

$${}^A T_B = \begin{pmatrix} 0 & -1 & 0 & 4 \\ 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(b) Using $T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$:

$$R^T = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad -R^T \vec{p} = - \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \\ 0 \end{pmatrix} = - \begin{pmatrix} 3 \\ -4 \\ 0 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \\ 0 \end{pmatrix}$$

$$({}^A T_B)^{-1} = \begin{pmatrix} 0 & 1 & 0 & -3 \\ -1 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

B2. Given the transformation matrix:

$${}^A T_B = \begin{pmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

A point P is at $(1, 2, 3)$ in frame $\{B\}$. Find the coordinates of P in frame $\{A\}$. **[5 marks]**

Solution B2

Express P in homogeneous coordinates: ${}^B P = (1, 2, 3, 1)^T$.

$${}^A P = {}^A T_B \cdot {}^B P = \begin{pmatrix} 0 & -1 & 0 & 3 \\ 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \\ 1 \end{pmatrix}$$

Computing row by row:

$$x_A = 0(1) + (-1)(2) + 0(3) + 3(1) = 0 - 2 + 0 + 3 = 1$$

$$y_A = 1(1) + 0(2) + 0(3) + 5(1) = 1 + 0 + 0 + 5 = 6$$

$$z_A = 0(1) + 0(2) + 1(3) + 2(1) = 0 + 0 + 3 + 2 = 5$$

$${}^A P = (1, 6, 5)$$

B3. Frame $\{C\}$ is obtained from $\{B\}$ by rotating 90° about the X -axis and translating $(0, 3, 0)$. Frame $\{B\}$ is obtained from $\{A\}$ by rotating 60° about the Z -axis and translating $(2, 0, 0)$. Find ${}^A T_C = {}^A T_B \cdot {}^B T_C$. **[6 marks]**

Solution B3**Step 1 — Build ${}^A T_B$ ($R_z(60^\circ)$, translation $(2, 0, 0)$):**

$${}^A T_B = \begin{pmatrix} 0.5 & -0.866 & 0 & 2 \\ 0.866 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Step 2 — Build ${}^B T_C$ ($R_x(90^\circ)$, translation $(0, 3, 0)$):

$$R_x(90^\circ) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad {}^B T_C = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Step 3 — Multiply:

$${}^A T_C = {}^A T_B \cdot {}^B T_C = \begin{pmatrix} 0.5 & -0.866 & 0 & 2 \\ 0.866 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Row 1: $(0.5)(1) + (-0.866)(0) = 0.5$; $(0.5)(0) + (-0.866)(0) + (0)(1) = 0$; $(0.5)(0) + (-0.866)(-1) = 0.866$; trans: $(0.5)(0) + (-0.866)(3) + 2 = -2.598 + 2 = -0.598$

Row 2: $(0.866)(1) + (0.5)(0) = 0.866$; $(0.866)(0) + (0.5)(0) = 0$; $(0.866)(0) + (0.5)(-1) = -0.5$; trans: $(0.5)(3) + 0 = 1.5$

Row 3: $(0)(1) + (0)(0) + (1)(0) = 0$; $(0)(0) + (0)(0) + (1)(1) = 1$; $(0)(0) + (0)(-1) + (1)(0) = 0$; trans: 0

$${}^A T_C = \begin{pmatrix} 0.5 & 0 & 0.866 & -0.598 \\ 0.866 & 0 & -0.5 & 1.5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

B4. A 2-link planar robot has Link 1: $L_1 = 4$, $\theta_1 = 30^\circ$ and Link 2: $L_2 = 3$, $\theta_2 = 45^\circ$. Using the DH planar HTM formula, find the transformation matrices ${}^0 T_1$ and ${}^1 T_2$, and determine the (x, y) position of the end-effector. [7 marks]

Solution B4

DH matrix for a planar link: ${}^{i-1} T_i = \begin{pmatrix} \cos \theta_i & -\sin \theta_i & 0 & L_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i & 0 & L_i \sin \theta_i \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$

${}^0 T_1$ ($\theta_1 = 30^\circ$, $L_1 = 4$, $\cos 30^\circ = 0.866$, $\sin 30^\circ = 0.5$):

$${}^0 T_1 = \begin{pmatrix} 0.866 & -0.5 & 0 & 3.464 \\ 0.5 & 0.866 & 0 & 2.0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

1T_2 ($\theta_2 = 45^\circ$, $L_2 = 3$, $\cos 45^\circ = 0.707$, $\sin 45^\circ = 0.707$):

$${}^1T_2 = \begin{pmatrix} 0.707 & -0.707 & 0 & 2.121 \\ 0.707 & 0.707 & 0 & 2.121 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

End-effector position using FK formulas directly:

$$x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) = 4 \cos 30^\circ + 3 \cos 75^\circ$$

$$= 4(0.866) + 3(0.2588) = 3.464 + 0.776 = \boxed{4.24 \text{ m}}$$

$$y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2) = 4 \sin 30^\circ + 3 \sin 75^\circ$$

$$= 4(0.5) + 3(0.9659) = 2.0 + 2.898 = \boxed{4.90 \text{ m}}$$

End-effector at $\approx (4.24, 4.90)$.

B5. The end-effector of a 2-DOF planar robot ($L_1 = 4$, $L_2 = 3$) must reach position (5, 5).

(a) Find θ_2 (both solutions: elbow-up and elbow-down). [3]

(b) Find θ_1 for the elbow-up solution. [2]

[5 marks]

Solution B5

(a) Using the IK cosine formula:

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2} = \frac{25 + 25 - 16 - 9}{2(4)(3)} = \frac{25}{24} \approx 1.0417$$

Since $\cos \theta_2 > 1$, this point is **outside the workspace** (maximum reach = $L_1 + L_2 = 7$, distance = $\sqrt{50} \approx 7.07 > 7$). The point is just barely unreachable.

If the question intends $(x, y) = (5, 4)$:

$$\cos \theta_2 = \frac{25 + 16 - 16 - 9}{24} = \frac{16}{24} = 0.667 \implies \theta_2 = \pm 48.19^\circ$$

$\theta_2 = +48.19^\circ$ (elbow-down); $\theta_2 = -48.19^\circ$ (elbow-up).

(b) For elbow-up ($\theta_2 = -48.19^\circ$):

$$k_1 = L_1 + L_2 \cos \theta_2 = 4 + 3(0.667) = 6.0, \quad k_2 = L_2 \sin \theta_2 = 3(-0.748) = -2.244$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(k_2, k_1) = \text{atan2}(4, 5) - \text{atan2}(-2.244, 6.0)$$

$$= 38.66^\circ - (-20.5^\circ) = \boxed{59.2^\circ}$$

SECTION C — Jacobian, Singularities & Force Statics [20 Marks]

C1. For the 2-DOF planar robot ($x = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$, $y = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$):

- (a) Derive the 2×2 Jacobian $J(\theta)$. [5]
- (b) The end-effector must move with $\dot{x} = 1$ m/s, $\dot{y} = 1$ m/s when $\theta_1 = 0$, $\theta_2 = 90^\circ$, $L_1 = L_2 = 0.5$ m. Find $\dot{\theta}_1$ and $\dot{\theta}_2$. [4]
- (c) Give two joint angle combinations (θ_1, θ_2) where dexterity is zero and explain why. [3]

[12 marks]

Solution C1**(a) Jacobian derivation:**Let $c_{12} = \cos(\theta_1 + \theta_2)$, $s_{12} = \sin(\theta_1 + \theta_2)$.

$$\frac{\partial x}{\partial \theta_1} = -L_1 \sin \theta_1 - L_2 \sin(\theta_1 + \theta_2) = -L_1 s_1 - L_2 s_{12}$$

$$\frac{\partial x}{\partial \theta_2} = -L_2 \sin(\theta_1 + \theta_2) = -L_2 s_{12}$$

$$\frac{\partial y}{\partial \theta_1} = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2) = L_1 c_1 + L_2 c_{12}$$

$$\frac{\partial y}{\partial \theta_2} = L_2 \cos(\theta_1 + \theta_2) = L_2 c_{12}$$

$$J(\theta) = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{pmatrix}$$

(b) At $\theta_1 = 0$, $\theta_2 = 90^\circ$: $\theta_1 + \theta_2 = 90^\circ$, $L_1 = L_2 = 0.5$.

$$s_1 = 0, c_1 = 1, s_{12} = 1, c_{12} = 0$$

$$J = \begin{pmatrix} -0(0.5) - 0.5(1) & -0.5(1) \\ 0.5(1) + 0.5(0) & 0.5(0) \end{pmatrix} = \begin{pmatrix} -0.5 & -0.5 \\ 0.5 & 0 \end{pmatrix}$$

 $\det(J) = (-0.5)(0) - (-0.5)(0.5) = 0.25 \neq 0$ (not singular).

$$J^{-1} = \frac{1}{0.25} \begin{pmatrix} 0 & 0.5 \\ -0.5 & -0.5 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ -2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{pmatrix} = J^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0(1) + 2(1) \\ -2(1) + (-2)(1) \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \end{pmatrix} \text{ rad/s}$$

(c) **Singularities** ($\det J = 0$): $\det J = L_1 L_2 \sin \theta_2$. So:

- $(\theta_1, 0^\circ)$ for any θ_1 : Arm fully extended — *boundary singularity*. Two links aligned; end-effector cannot move radially.
- $(\theta_1, 180^\circ)$ for any θ_1 : Arm fully folded back on itself — another *boundary singularity*.

C2. A robotic arm's Jacobian at a configuration is:

$$J = \begin{pmatrix} 0.5 & -0.3 \\ 0.866 & 0.5 \end{pmatrix}$$

An external force $\vec{F} = (10, 5)^T$ N is applied at the end-effector. Using the static force relation $\vec{\tau} = J^T \vec{F}$, find the required joint torques. Also verify that the virtual work principle is satisfied. [8 marks]

Solution C2**Step 1 — Compute J^T :**

$$J^T = \begin{pmatrix} 0.5 & 0.866 \\ -0.3 & 0.5 \end{pmatrix}$$

Step 2 — Compute $\vec{\tau} = J^T \vec{F}$:

$$\begin{pmatrix} \tau_1 \\ \tau_2 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.866 \\ -0.3 & 0.5 \end{pmatrix} \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 0.5(10) + 0.866(5) \\ -0.3(10) + 0.5(5) \end{pmatrix} = \begin{pmatrix} 5 + 4.33 \\ -3 + 2.5 \end{pmatrix}$$

$$\boxed{\vec{\tau} = \begin{pmatrix} 9.33 \text{ Nm} \\ -0.5 \text{ Nm} \end{pmatrix}}$$

Virtual work verification:For virtual displacements $\delta \vec{q} = (0.1, 0.2)^T$ rad:

$$\delta W_{\text{joint}} = \vec{\tau}^T \delta \vec{q} = 9.33(0.1) + (-0.5)(0.2) = 0.933 - 0.1 = 0.833 \text{ J}$$

$$\delta \vec{x} = J \delta \vec{q} = \begin{pmatrix} 0.5 & -0.3 \\ 0.866 & 0.5 \end{pmatrix} \begin{pmatrix} 0.1 \\ 0.2 \end{pmatrix} = \begin{pmatrix} 0.05 - 0.06 \\ 0.0866 + 0.1 \end{pmatrix} = \begin{pmatrix} -0.01 \\ 0.1866 \end{pmatrix}$$

$$\delta W_{\text{tip}} = \vec{F}^T \delta \vec{x} = 10(-0.01) + 5(0.1866) = -0.1 + 0.933 = 0.833 \text{ J} \quad \checkmark$$

Both virtual works are equal, confirming $\vec{\tau} = J^T \vec{F}$.**SECTION D — DC Motors, Circuits & Control [30 Marks]**

D1. A DC motor operates with $V = 80 \text{ V}$, $R = 4 \Omega$, $K = 2 \text{ Vs/rad}$, constant load torque $T_L = 10 \text{ Nm}$. At steady state $T_e = KI$ and $V = IR + K\omega$. Find the steady-state speed ω . [6 marks]

Solution D1**Step 1 — Find steady-state current from torque balance ($T_e = T_L$):**

$$T_e = KI \implies I = \frac{T_L}{K} = \frac{10}{2} = 5 \text{ A}$$

Step 2 — Apply motor voltage equation:

$$V = IR + K\omega \implies \omega = \frac{V - IR}{K} = \frac{80 - (5)(4)}{2} = \frac{80 - 20}{2} = \frac{60}{2}$$

$$\boxed{\omega = 30 \text{ rad/s}}$$

D2. A motor ($V = 24 \text{ V}$, $R = 2 \Omega$, $K = 0.5 \text{ Nm/A}$) drives a viscous load $T_L = B\omega$ with $B = 0.1 \text{ Nms/rad}$. At steady state $T_e = T_L$. Find the operating speed. [6 marks]

Solution D2

At steady state: $T_e = T_L$, so $KI = B\omega \implies I = \frac{B\omega}{K}$.
 Substituting into the voltage equation:

$$V = IR + K\omega = \frac{B\omega}{K} \cdot R + K\omega = \omega \left(\frac{BR}{K} + K \right)$$

$$\omega = \frac{V}{\frac{BR}{K} + K} = \frac{VK}{BR + K^2} = \frac{24 \times 0.5}{(0.1)(2) + (0.5)^2} = \frac{12}{0.2 + 0.25} = \frac{12}{0.45}$$

$\omega = 26.67 \text{ rad/s}$

D3. An elevator hoist motor ($K = 2.0 \text{ Nm/A}$, $R = 1 \Omega$, $V = 100 \text{ V}$) drives a constant gravitational load $T_L = 10 \text{ Nm}$, regardless of speed. Find the steady-state operating speed.
[5 marks]

Solution D3

Steady-state current (from torque balance, $T_e = T_L$):

$$I = \frac{T_L}{K} = \frac{10}{2} = 5 \text{ A}$$

Speed from voltage equation:

$$\omega = \frac{V - IR}{K} = \frac{100 - (5)(1)}{2} = \frac{95}{2} = \boxed{47.5 \text{ rad/s}}$$

D4. A robotic wheel motor ($V = 48 \text{ V}$, $R = 1.5 \Omega$, $K = 0.8 \text{ Nm/A}$) runs at steady state under $T_L = 6 \text{ Nm}$. The supply drops suddenly to 36 V while load torque is unchanged. Find the new steady-state speed.
[5 marks]

Solution D4

Since T_L is unchanged, the steady-state current is the same:

$$I = \frac{T_L}{K} = \frac{6}{0.8} = 7.5 \text{ A}$$

New speed at $V = 36 \text{ V}$:

$$\omega_{\text{new}} = \frac{V_{\text{new}} - IR}{K} = \frac{36 - (7.5)(1.5)}{0.8} = \frac{36 - 11.25}{0.8} = \frac{24.75}{0.8}$$

$\omega_{\text{new}} = 30.94 \text{ rad/s}$

(Compare to original: $\omega_{\text{old}} = \frac{48 - 11.25}{0.8} = 45.94 \text{ rad/s}$ — voltage drop reduced speed by 33%.)

D5. A motor requiring an average DC voltage of 30 V is driven by an H-bridge connected to a 150 V DC supply switching at 10 kHz .

(a) Determine the duty cycle D and the ON-time t_{on} per switching cycle. [3]

- (b) During t_{on} , the MOSFET has $V_G = 12 \text{ V}$, $V_{TH} = 3 \text{ V}$, $V_D = 7 \text{ V}$. Identify the operating region. [2]

[5 marks]

Solution D5

(a) $V_{dc} = D \cdot V_s \implies D = \frac{V_{dc}}{V_s} = \frac{30}{150} = \boxed{0.2 = 20\%}$

Switching period: $T = \frac{1}{f} = \frac{1}{10000} = 100 \mu\text{s}$

$t_{\text{on}} = D \times T = 0.2 \times 100 \mu\text{s} = \boxed{20 \mu\text{s}}$

(b) Check MOSFET region:

$$V_{GS} \approx V_G = 12 \text{ V} \quad (\text{since source at ground})$$

$$V_{GS} - V_{TH} = 12 - 3 = 9 \text{ V}$$

$$V_{DS} = V_D - V_S = 7 - 0 = 7 \text{ V}$$

Since $V_{DS} = 7 \text{ V} < V_{GS} - V_{TH} = 9 \text{ V}$:

MOSFET is in the **Triode (Ohmic) region**

This is the desired conducting state for a motor drive switch (low V_{DS} , low losses).

- D6.** During H-bridge testing on a 240 V DC supply, a shoot-through fault creates a short with parasitic resistance $R = 0.08 \Omega$ and parasitic inductance $L = 12 \mu\text{H}$. Back EMF is zero and initial current $i(0) = 0$. Determine the instantaneous rate of change of current $\left(\frac{di}{dt}\right)$ at $t = 0^+$. [3 marks]

Solution D6

At $t = 0^+$, current $i = 0$ so the resistive drop is negligible. By the inductor relation $V = L \frac{di}{dt}$, the full supply voltage appears across the parasitic inductance:

$$\left. \frac{di}{dt} \right|_{t=0^+} = \frac{V}{L} = \frac{240}{12 \times 10^{-6}}$$

$\left. \frac{di}{dt} \right|_{t=0^+} = 20,000,000 \text{ A/s} = 20 \text{ MA/s}$

This catastrophically large rate of current rise is precisely why shoot-through faults destroy power electronics. The current rises nearly without limit until a protection circuit or device fails.

HEET — Introduction to Robotics & Intelligent Control Systems

This sheet may be detached and used during examination. No additional formulae allowed.

Unit 1: Rotation Matrices

$$R_x(\phi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi & c_\phi \end{pmatrix}$$

$$R_y(\theta) = \begin{pmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{pmatrix}$$

$$R_z(\psi) = \begin{pmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Properties: $R^{-1} = R^T$, $\det(R) = +1$
YPR: $R = R_z(\psi)R_y(\theta)R_x(\phi)$

Unit 1: HTM

$$T = \begin{pmatrix} R_{3 \times 3} & \vec{p}_{3 \times 1} \\ \mathbf{0} & 1 \end{pmatrix}, \quad T^{-1} = \begin{pmatrix} R^T & -R^T \vec{p} \\ \mathbf{0} & 1 \end{pmatrix}$$

Chaining: ${}^0T_n = {}^0T_1 \dots {}^{n-1}T_n$
Point transform: ${}^A\vec{p} = {}^AT_B \cdot {}^B\vec{p}$

Unit 2: DH Parameters

θ_i	Joint angle (Z_{i-1})	Variable (revolute)
d_i	Offset (Z_{i-1})	Variable (prismatic)
a_i	Link length (X_i)	Fixed
α_i	Twist (X_i)	Fixed

$${}^{i-1}T_i = \begin{pmatrix} c\theta_i & -s\theta_i c\alpha_i & s\theta_i s\alpha_i & a_i c\theta_i \\ s\theta_i & c\theta_i c\alpha_i & -c\theta_i s\alpha_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Planar: $d_i = 0, \alpha_i = 0$

Unit 3: Inverse Kinematics (2-DOF)

$$\cos \theta_2 = \frac{x^2 + y^2 - L_1^2 - L_2^2}{2L_1L_2}$$

$\theta_2 = \pm \arccos(\cdot)$ (elbow-up/down)

$$k_1 = L_1 + L_2 c_2, \quad k_2 = L_2 s_2$$

$$\theta_1 = \text{atan2}(y, x) - \text{atan2}(k_2, k_1)$$

Workspace: $|L_1 - L_2| \leq r \leq L_1 + L_2$

Unit 4: Jacobian

$$\dot{\vec{x}} = J(\vec{q})\dot{\vec{q}}$$

2-DOF Planar:

$$J = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} & -L_2 s_{12} \\ L_1 c_1 + L_2 c_{12} & L_2 c_{12} \end{pmatrix}$$

$$\det(J) = L_1 L_2 \sin \theta_2$$

Singular when $\theta_2 = 0^\circ$ or 180°

$$\text{Dexterity: } w = \sqrt{\det(JJ^T)} = |\det J|$$

Unit 5: Force & Statics

$$\vec{\tau} = J^T \vec{F}$$

Virtual work: $\vec{\tau}^T \delta \vec{q} = \vec{F}^T \delta \vec{x}$

$$\delta \vec{x} = J \delta \vec{q}$$

Unit 6: DC Motor

Voltage equation: $V = IR + K\omega + L \frac{di}{dt}$
(Steady state: $V = IR + K\omega$)

Torque: $T_e = KI = K_t I$

Back EMF: $E_b = K_e \omega$

Steady-state speed:

$$\omega = \frac{V - IR}{K} = \frac{V - T_L R / K}{K} = \frac{VK - T_L R}{K^2}$$

Stall torque ($\omega = 0$): $I = V/R$, $T = KV/R$

No-load speed ($I \approx 0$): $\omega_0 = V/K$

Viscous load ($T_L = B\omega$): $\omega = \frac{VK}{BR + K^2}$

Power: $P = \tau\omega = VI \cdot \eta$

Unit 7: Circuits & Control

Ohm's Law: $V = IR$, $P = VI = I^2 R = V^2 / R$

KCL: $\sum I_{\text{in}} = \sum I_{\text{out}}$

KVL: $\sum V = 0$ (around any loop)

Series: $R_{eq} = R_1 + R_2 + \dots$

Parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

PWM/H-bridge: $V_{dc} = D \cdot V_s$, $T = 1/f$,
 $t_{\text{on}} = D \cdot T$

MOSFET regions:

Triode: $V_{DS} < V_{GS} - V_{TH}$

Saturation: $V_{DS} \geq V_{GS} - V_{TH}$

Inductor: $V = L \frac{di}{dt}, \frac{di}{dt} \Big|_{t=0+} = \frac{V}{L}$

Shoot-through (H-bridge): all voltage on L

Half-wave rectifier: $V_{dc} = \frac{V_m}{\pi}$

Useful Trig Values: $\cos 30^\circ = \sin 60^\circ = \frac{\sqrt{3}}{2} \approx 0.866$ $\cos 45^\circ = \sin 45^\circ = \frac{1}{\sqrt{2}} \approx 0.707$
 $\cos 60^\circ = \sin 30^\circ = 0.5$ $\cos 90^\circ = 0, \sin 90^\circ = 1$

$\text{atan2}(y, x)$: angle of vector (x, y) in range $(-180^\circ, 180^\circ]$. 2×2 inverse:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$